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EDITORIAL

Overview of Best Practices for Biopharmaceutical Technology Transfers

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ABSTRACT: Technology transfer is a key foundational component in product commercialization. It is more than just the transfer of documents; it relates to all aspects of the transfer of knowledge and experience to the commercial manufacturing unit to ensure consistent, safe, and high-quality product. This is the first in a series of articles from the BioPhorum Operations Group (BPOG) member companies discussing best practices and benchmarking of biopharmaceutical technology transfer. In this article, we provide the common terminology developed by BPOG to accommodate both transferring and receiving organizations. We also review the key elements of a robust technology transfer business process, including critical milestones. Finally, we provide a brief overview of the articles in this series.

Introduction

Technology transfer (TT)—the transfer of product and process knowledge between development and manufacturing, and within or between manufacturing sites, to achieve product realization (ICH Q10)—is a critical business process in the biopharmaceutical industry. As processes are scaled up for commercial manufacturing or expanded to alternative sites for expansion or business risk mitigation, the effectiveness of TT can have a substantial effect on the realization of business goals. Given the highly technical nature of biopharmaceuticals and the professionals working in it, it is no surprise that we are quite good at TT from a technical perspective. However, companies feel that we could do it faster and more cost effectively to increase returns on expensive biopharmaceutical development and approval activities.

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Understanding how to streamline and improve TTs is complicated by companies using different terminology and ways of working. This in turn makes assessments and meaningful metrics difficult to define. When, for example, does a TT begin—when a steering committee is formed, a project charter is approved, or when the receiving site is selected? Similarly the definition of the end of a TT was diverse between organizations, with variation between milestones such as the end of the campaign, lot disposition, regulatory submission, or regulatory dossier approval. It rapidly became clear to us that a common terminology and language was need for the sharing of current and future best practices. The sharing of the terminology, best practices, and of benchmarking within the BioPhorum Operations Group (BPOG) Technology Transfer Working Group is the goal of this series of articles.

Figure 1 illustrates the key quality, technical, and business drivers that need to be balanced to ensure the success of TTs. TT is considered successful when we get it *"right first time"* with successful lot release and a minimal number of non-conformances or out-of-specification events. From a technical perspective, it is necessary to demonstrate that the process scale-up and transfer is effective by analyzing key and critical process performance parameters and achieving targets

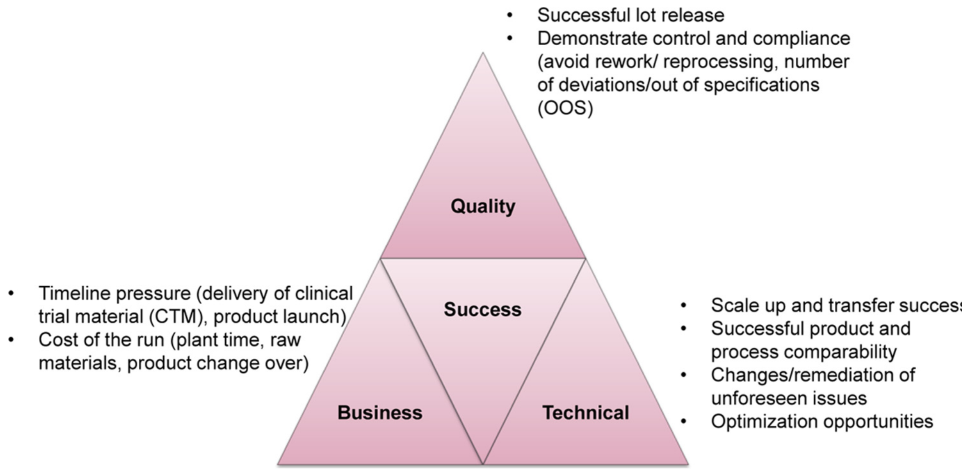

Figure 1

Key drivers for successful technology transfer (TT).

for product quality attributes and process yield. The quality and technical drivers need to be achieved against the backdrop of business constraints. These include schedule and on-time delivery of product for clinical trials and product launch, and financial and resource considerations.

Definition of Terms

As noted in the introduction, it is important to ensure a common framework and terminology for a meaningful discussion. The TT framework discussed here refers to the transfer of biopharmaceutical processes from the sending unit to the receiving unit, the latter typically being a commercial manufacturing operation. This involves both process transfer and analytical transfer (method qualification, transfer, robustness, and validation or verification). The BPOG Technology Transfer Working Group definitions are summarized in Table I.

Key Activities and Milestones In Technology Transfer (TT) Activities

Once a set of terms were defined, we were able to discuss and compare the actual business process flow for TT. Although not identical, the prerequisites, responsible units, and the type of output for a given step in successful TTs were found to be highly similar. The activity flow for TTs (Table II), based on the elements of successful transfers, was developed. This flow was similar between large companies and small companies, sending units (SUs) and receiving units (RUs), and between product sponsors/clients and Contract Manufacturing Organizations (CMOs). It is the basis of the TT business process outlined with a focus on late stage—Phase 3 clinical and process performance qualification (PPQ)—and post-licensure TTs. Many of the elements also apply to early-stage transfers. In addition, we identified key milestones for the TT, and these are summarized in Table II.

Articles In the Series

This is the first of a series of articles to reflect the current and best practices, as well as learnings from participants in the forum. With an industry-wide perspective, improvements can be made in the TT process that we trust will benefit the industry as a whole. Some of the topics addressed in later articles will include shakedown and engineering runs, and risk management in TT.

Conclusion

Using common terminology and language, current and future best practices can be shared across the industry. This article has outlined the necessary elements of business process flow for TTs that are highly similar throughout the biopharmaceutical industry. The comparisons and development of industry-wide best practices will provide greater efficiencies from improved ways of working. Once in place,

TABLE I
Definition of Terms

Term	Definition
Analytical Method Qualification	A documented process or partial validation designed to verify the method is suitable for its intended purpose. It is suitable for early investigational new drug (IND) studies and characterization assays, as it assesses a critical subset of validation characteristics.
Analytical Method Transfer	A documented process designed to verify a certain laboratory's capability of performing the analytical testing method's intended use.
Analytical Method Validation	A documented process designed to assess a method against all appropriate validation characteristics. See ICH Guideline: Q2 (R1): Validation of Analytical Procedures: Text and Methodology. It has pre-defined acceptance criteria, and it follows formal validation protocol/sign off by the quality Unit.
Change Request	Good manufacturing practice (GMP)-documented request approved by the quality unit, for modifications of processes; in the case for TT, this can apply to a site change, a scale change, or a process change required for facility fit. It may not be applicable to initial activities that are non-product-specific for transfer or for transfers of processes that have not been validated.
Clinical Lots	Full-scale lots manufactured to supply clinical studies. These can function as engineering lots for process performance qualification (PPQ) later in a development program.
Commercial Lots	Full-scale lots manufactured to supply material for its intended (therapeutic) use. These may include the process performance qualification (PPQ) and continuous process verification (CPV) lots and are targeted as product for commercial sale/distribution.
Comparability Studies	The demonstration that the quality attributes are highly similar and that the existing knowledge is sufficiently predictive to ensure that any differences in quality attributes have no adverse impact upon safety or efficacy of the drug product (ICH Q5E).
Dry Runs	Batch record (BR) walk down on the manufacturing shop floor to ensure that the BR record documentation aligns with manufacturing's ability to execute the process (transfer lines, sampling, sequence of operations, clarity of instructions). Synonym: batch record (BR) walk down.
Facility Fit	Determination of how the process will be executed at the receiving unit (RU) compared to the sending unit (SU). Captures all differences and/or gaps for end-to-end process execution. This includes facilities and equipment, cleaning and sterilization procedures, environmental health and safety procedures, raw materials and components handling, quality control (QC) and analytical activities, and quality systems.
Engineering Lots	Full-scale lots of the final process to demonstrate process performance and quality attributes at the receiving unit (RU) prior to process performance qualification (PPQ). These may be GMPs for the manufacture of clinical or commercial material. For clinical manufacturing, the engineering lots may be the clinical manufacturing lots. Synonyms: first full-scale clinical (FSC) lot, first clinical/commercial-scale lot (FCS).
Gap Analysis	Identification of risks for the process at the receiving unit (RU). Captures all differences and/or gaps in knowledge at the RU for end-to-end process execution. This includes facilities and equipment, cleaning and sterilization procedures, environmental health and safety procedures, raw materials and components handling, quality control (QC) and analytical activities, and quality systems. Synonyms: quality risk assessments (QRAs).
Good Manufacturing Practice (GMP) Campaign	Full-scale lots manufactured to supply material for its intended (therapeutic) use. These can be clinical batches, process performance qualification (PPQ) batches, or commercial supply batches.
Feasibility Evaluation	Identification of major elements of a process which are different at the receiving unit (RU) compared to the sending unit (SU), that could affect transfer timing or feasibility including facilities and equipment, key procedures, raw materials and components handling, quality control (QC) and analytical support, and quality systems. Synonyms: technical evaluation, high-level gap analysis.
High-Level Process Documentation Package	Generalized documents describing the process for the sending unit (SU) and receiving unit (RU). They do not include all details, but only sufficient information, for the high-level gap analysis and high-level TT plan.
High-Level Technology Transfer Plan	A site-specific but general plan for TT activities to assess major activities, investments, and feasibility. Synonym: high-level project plan.
Laboratory Demonstration	Demonstration runs, performed by the receiving unit (RU) and based on a predefined plan and acceptance criteria, showing the performance of the process or analytical methods at laboratory scale. This occurs prior to scale-up or method qualification. Synonym: method familiarization (typically performed by the RU).
Process Documentation Package	Documents describing the process for the sending unit (SU) and receiving unit (RU). May contain standard operating procedures (SOPs), master batch records (MBRs), process descriptions (PDs), development reports, validation reports, engineering drawings, material lists, sampling plan, etc. Synonym: technical documentation package (TDP).

TABLE I (Continued)

Term	Definition
Process Performance Qualification (PPQ)	Confirming that the manufacturing process as designed is capable of reproducible commercial manufacturing (FDA Guidance for Industry, "Process Validation: General Principles and Practices", January 2011, Revision 1).
Process Robustness	Ability of a process to tolerate variability of materials and changes of the process and equipment without negative impact on quality or productivity.
Process Transfer	The transfer of process and analytical knowledge between development and manufacturing, or between manufacturing sites, to establish the process at the receiving unit (RU). Process transfer is a subset of the activities for TT.
Process Transfer Completion	As defined in a study plan, successful disposition (release) of material from initial batch(es) by the receiving unit (RU).
Process Transfer Initiation	The providing of process and analytical knowledge by the sending unit (SU), usually in the form of the process documentation package, and commencement of activity at the receiving unit (RU) for detailed gap analysis and laboratory demonstration.
Process Transfer Success Criteria	Pre-defined criteria for assessing the success or effectiveness of the process. These could be considered the key performance indicators (KPIs) of the process. Examples could be meeting drug substance (DS) specification and in process control action limits, comparability criteria, meeting drug substance/drug product (DS/DP) release dates, or no major rework or reprocessing required.
Project Charter	Document describing of the goals of the TT, including scope of process to be transferred, definition of the transferring and receiving sites, project team members and structure, governance structure, major milestones/deliverables, assumptions, risks, business benefits, and overall acceptable outcome.
Project Team	A team, having members with clearly defined key responsibilities, drawn from members of relevant disciplines from both the sending unit (SU) and receiving unit (RU) sites, to manage the TT. Synonym: core project team.
Receiving Unit (RU)	The involved disciplines at an organization to where a designated product, process, or method is expected to be transferred.
Sending Unit (SU)	The involved disciplines at an organization from where a designated product, process, or method is expected to be transferred.
Shakedown Lots	Experimental batches for execution of the process at scale for development of the final manufacturing process, automation, and operation. Shakedown runs are experimental only, and are not used for clinical or commercial supply manufacturing. The shake-down run includes use of cell culture or product containing feed streams. Synonym: not for human use (NfHU) run.
Stability Studies	Testing to provide evidence on how the quality of a drug substance, drug product, or placebo varies with time under the influence of a variety of environmental factors, such as temperature, humidity, and light, and to establish a retest period for the drug substance or a shelf life for the drug product and recommended storage conditions.
Steering Committee	Team responsible for oversight/direction of the TT project/core team. The TT steering committee is typically composed of senior management members from the sending unit (SU) and receiving unit (RU) as well as additional functions as needed for critical oversight of the TT. Synonyms: TT oversight committee.
Sub-Team	A team, having members with clearly defined key responsibilities, with members representing the discipline from both the sending unit (SU) and receiving unit (RU), to oversee that disciplines activities as part of the TT. Examples include analytical, process, upstream, downstream, raw materials, production, regulatory affairs, quality control (QC), engineering, supply chain, etc. Synonym: workstream.
Technology Transfer (TT)	The transfer of product and process knowledge between development and manufacturing, and within or between manufacturing sites to achieve product realization (ICH Q10).
Technology Transfer Completion	License/facility approval for manufacturing by the receiving unit (RU) of the transferred process, and demonstration of robust manufacturing.
Technology Transfer Initiation	Formal selection of the receiving unit (RU).
Technology Transfer Plan	A document, approved by the sending unit (SU) and receiving unit (RU), defining the task assignments and responsibilities, acceptance criteria for gates and the completion of TT, documentation to be generated, and timing of activities. Synonym: technology transfer master plan (TTMP).
Technology Transfer Success Criteria	Pre-defined evaluation criteria for assessing the success or effectiveness of the TT process. These would include success of the process transfer, demonstration of robustness of the operation at the receiving unit, and license/facility approval for manufacturing.

TABLE II
Activity Flow for Technology Transfers (TTs)

Stage	Sub-Stages	Milestones	Gate Requirements
TT Start		• Feasibility evaluation complete • Receiving site identified • Services and quality agreement established (as needed) • Steering committee established • Project team assembled with membership from the SU and RU • Defined TT success criteria	• Project charter • High-level project plan • Budget approved • High-level TT plan and schedule
Process Transfer Initiation	**Information Exchange**	• Process documentation package assembled • Define regulatory strategy • Initiate facility fitu • Define and initiate analytical method transfer for required methods	• Complete and accepted process documentation package • Analytical documentation package
	Gap Analysis	• Facility fit for the process in the RU • Initiate change request	• Facility fit report • Gap analysis and remediation plan
Process Transfer	**Planning**	• Creation of detailed individual workstream TT plans (e.g., process, analytical methods, materials, equipment, comparability, etc.) with costs and resources • Update gap analysis and risk assessments	• Detailed TT plans and schedules
	Execution	• Laboratory demonstration • Studies to fill knowledge gaps and to support changes • PPQ/campaign preparation ○ Training ○ Procedures ○ Document generation/verification ○ Equipment qualification ○ Raw material release • Analytical method qualification • Dry runs, shakedown lots • Engineering lots	• Process, documentation, and facility readiness for PPQ or initial campaign.
Process Transfer Completion	**GMP Campaign**	• Analytical method validation • Successful PPQ runs or initial batches	• Validation or initial campaign summary report
TT Completion	**Submission and Licensure**	• Comparability studies reports complete • Stability studies reports complete • TT summary report • Assembly and submission of dossier • Pre-approval inspection (PAI)	• Successful pre-approval inspection (PAI) • Approval
	Post-Transfer	• Lessons learned report • Lessons learned communication • Creation of technical service support plan	• Communication and archival of lessons learned • Team celebration

these can result in reductions in timelines, costs, and regulatory risks.

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